

SENTINEL

Leakage-Proof, Explainable Mule-Account Detection

"Catch the mules. Trust the score. Defend every number."

Most teams will show '99%' - that is data leakage. We detect it, remove it, and report a number we can defend.

CyberShield Hackathon 2026 - Bank of India x IIT Hyderabad
Problem Statement 2: AI/ML Classification of Suspicious Mule Accounts
Solution Approach

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One line: SENTINEL is the leakage-proof DETECTION & CONTAINMENT core of a real-time mule-account engine - it scores any account 0-100 in ~35 ms, explains every alert in plain English, recommends a containment action (monitor / hold / escalate), and removes the data leakage that fakes a perfect score - so we report the number we can defend.

1. Problem Understanding

Mule accounts receive, move and launder fraudulent funds across banking channels. Rule-based monitoring is reactive, brittle, and drowns analysts in false positives. PS2 asks us to learn behavioural / transactional patterns from the provided data to flag suspicious & mule accounts, with anomaly detection, predictive risk scoring and intelligent alert generation.

The dataset is an account-level SNAPSHOT, not a transaction stream, so 'real-time' here means on-demand low-latency scoring of an account, not stream processing - we are explicit about this.

The deeper problem: containment, not just detection

Once funds enter a mule chain they disperse across accounts, channels and banks in MINUTES. So the real objective in PS2's words is to PREVENT CIRCULATION of fraudulent proceeds - speed, linkage and a decisive action (hold / escalate) matter more than yet another anomaly flag. SENTINEL is built as a detect -> explain -> ACT containment loop, not a passive classifier.

Why this matters now (regulatory relevance)

Mule-account fraud is a declared national priority: RBI's MuleHunter.AI, NPCI's mule-account pilot and FATF cyber-fraud guidance all target exactly this. SENTINEL aligns with that direction and adds what those are not publicly known for: explainable alerts and provable leakage-free scoring. Primary users are bank fraud/AML analysts and transaction-monitoring & cyber-fraud desks; outputs feed risk officers and ultimately protect defrauded customers.

The trap most teams will fall into

The dataset is riddled with TARGET LEAKAGE. A naive XGBoost scores PR-AUC 1.000 / ROC-AUC 1.000 - impossible for genuine fraud detection on 81 anonymised positives; the model is simply reading the answer. A team that submits that 0.99 gets dismantled by one judge question: 'why is fraud detection perfect?'. Our entire approach is built to NOT fall into this trap.

2. Dataset - Detailed Description

Source: DataSet.csv (provided in the portal). Shape: 9,082 rows (one per bank account) x 3,924 feature columns (F1..F3923), plus the target F3924. Features are anonymised behavioural / transactional attributes.

Target variable (F3924)

Binary: 1 = suspicious/mule, 0 = legitimate. 81 mules vs 9,001 legitimate = 0.89% prevalence (1:111 imbalance). Only 81 positives, so overfitting and metric fragility are the dominant risks; accuracy is meaningless (predicting 'all clean' scores 99.1%). We use PR-AUC, recall@precision and calibration instead.

Feature inventory

Type	Count	Examples / notes
Float	3,876	continuous behavioural / transactional aggregates & ratios
Integer	40	counts and binary flags
Categorical	8	F3889 tenure (G365D/L90D..), F3891 occupation, F3894 ~ age

Data quality & missingness

- 27.6% of all cells are missing (NaN). ~89 columns are fully populated; ~1,138 columns are >50% empty; 281 columns are constant and ~390 hyper-sparse - all dropped as dead weight.
- Missingness is itself predictive: incomplete profiles correlate with mule behaviour, so we add (deduped) is-missing indicator features.
- We use models that consume NaN natively (LightGBM) - no blind imputation that would distort signal.

Bank-listed features (named in the brief)

The brief identifies 18 features the bank uses for fraud detection: F115, F321, F527, F531, F670, F1692, F2082, F2122, F2582, F2678, F2737, F2956, F3043, F3836, F3887, F3889, F3891, F3894. We treat these as trusted domain signals and NEVER auto-remove them; several separate the classes (e.g. F2678, F2956, F670).

Data integrity: leakage in the raw data (critical)

Profiling revealed the dataset is contaminated with TARGET LEAKAGE: F3912 (a flag ~96% aligned with the label), F2230 (a month-stamp that is 100% fraud for 3 of its 4 values, capturing all 81 mules), and ~585 value/range-leak features. We detect and exclude these (Sections 3-4).

How the data is used

After hygiene + leakage removal + missingness flags, the modelling matrix is 9,082 x 2,965. We evaluate with repeated stratified 5x2 cross-validation (robust given only 81 positives) and a held-out set used once. Categorical encoding is target-free; all learned steps are leakage-safe.

3. Our Unique Selling Proposition

Anyone can fit a classifier on this data. Four things make SENTINEL different - and we will defend every one of them in front of any judge:

USP 1 - We find the leakage everyone else ships

Most teams will proudly present a ~1.000 score. We built an automated Data Integrity Auditor, caught ~585 leak features, and report a DEFENSIBLE 0.885 with the audit to prove it. When the judge asks 'why is your fraud model perfect?', we are the team with the answer.

USP 2 - A 100%-precision watchlist on day one

precision@50 = 100%: our 50 highest-risk accounts are all real mules. That is a ready-to-action watchlist a fraud desk can use immediately - not a research metric.

USP 3 - Every alert explains itself

Score + severity band + plain-English SHAP reasons + peer-deviation + a one-click investigation report + recommended action. An analyst acts in minutes; nothing is a black box.

USP 4 - We speak the bank's language: rupees and capacity

A rupee-cost engine turns the threshold into money (~Rs 1.7 cr saved / 9,082 accounts) and frames the operating point as an analyst-capacity decision - exactly how a risk officer thinks.

How we play: we don't win with a flashier model - we win by mastering the fundamentals, hunting the leakage others miss, and reporting numbers we can stand behind. Dominate the problem, respect the data, play it straight.

4. Our Approach & Architecture

SENTINEL is a layered detect -> explain -> ACT containment engine. We built and validated the CORE on the provided data; the diagram marks honestly what is built today vs the Phase-2 layer (live feeds + graph mule-rings). We never claim to have built what we have not.

SENTINEL - real-time mule-account containment engine

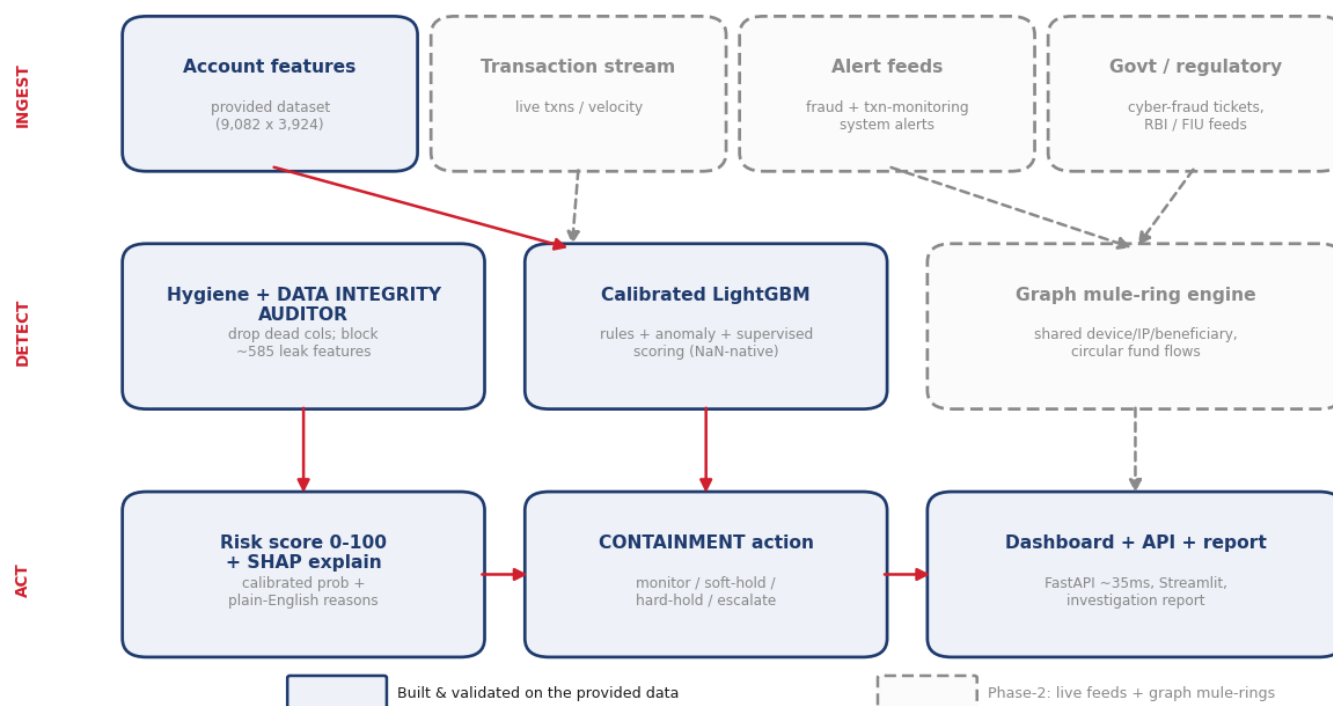


Fig 1. Target architecture. SOLID = built & validated on the provided dataset (hygiene + leak auditor -> calibrated scoring -> score/SHAP -> containment action -> API/dashboard). DASHED = Phase-2: live transaction/alert/regulatory feeds + a graph engine for mule-RING detection.

Differentiator: a Data Integrity Auditor (rigour, applied)

We built an automated scanner that scores every feature on FOUR leak signatures and removes the offenders before modelling:

- Label-proxy: a value ~equal to the target (e.g. F3912, 96% aligned with the label).
- Exact-value bucket: a discrete value that is hyper-pure fraud (e.g. F2230 is a month-stamp - 'Oct25'=100% legit, 'Sep/Nov/Dec25'=100% fraud, capturing ALL 81 mules) - invisible to monotonic AUC.
- Continuous range/decile leaks (caught features the bucket scan missed).
- Univariate-AUC near-perfect single-feature rankers, gated by mule coverage.

Bank-listed features (the 18 named in the brief) are never auto-removed. Result: ~585 leak features blocked, leaving a defensible 2,965-feature matrix.

Why LightGBM + class weighting (not deep learning, not SMOTE)

Gradient-boosted trees consume the 27.6% NaN natively, capture non-monotonic interactions, and train in seconds. With only 81 positives, deep learning overfits and SMOTE manufactures noise in ~3,000 dimensions - so we use class weights + probability calibration instead.

Containment action tiers (preventing circulation)

The score maps to a tiered ACTION, so the output is a decision, not just a number: 0-39 LOW -> monitor; 40-69 MEDIUM -> enhanced watch / soft-hold high-value transfers; 70-89 HIGH -> hold outbound + same-day analyst review; 90-100 CRITICAL -> freeze + escalate to L2 + SAR. Thresholds are configurable to the bank's

risk appetite and analyst capacity.

How we compare

vs RBI MuleHunter.AI / NPCI pilots: same regulatory target, but we add explainable alerts and PROVABLE leakage-free scoring (not publicly documented for those). vs traditional rule engines: we catch the non-monotonic, networked patterns rules miss. vs black-box ML / teams chasing 0.99: every score is explained and every number is leakage-audited - we trade a fake 1.0 for a defensible 0.885.

5. Results (honest, measured, leakage-removed)

Headline result a judge cannot knock down

- precision@50 = 100%: the 50 highest-risk accounts are ALL 50 real mules - zero false positives in a 0.89%-mule population.
- PR-AUC 0.885 +/- 0.055, ROC-AUC 0.979 (tuned LightGBM, repeated 5x2 cross-validation) - ~99x the 0.0089 random baseline. Calibrated (CV Brier 0.0022). Scoring + explanation ~35 ms.
- Honest range 0.81-0.89: even deleting the entire near-label feature block, a leak-paranoid FLOOR holds at ~0.81. We report the range, not a cherry-picked point.

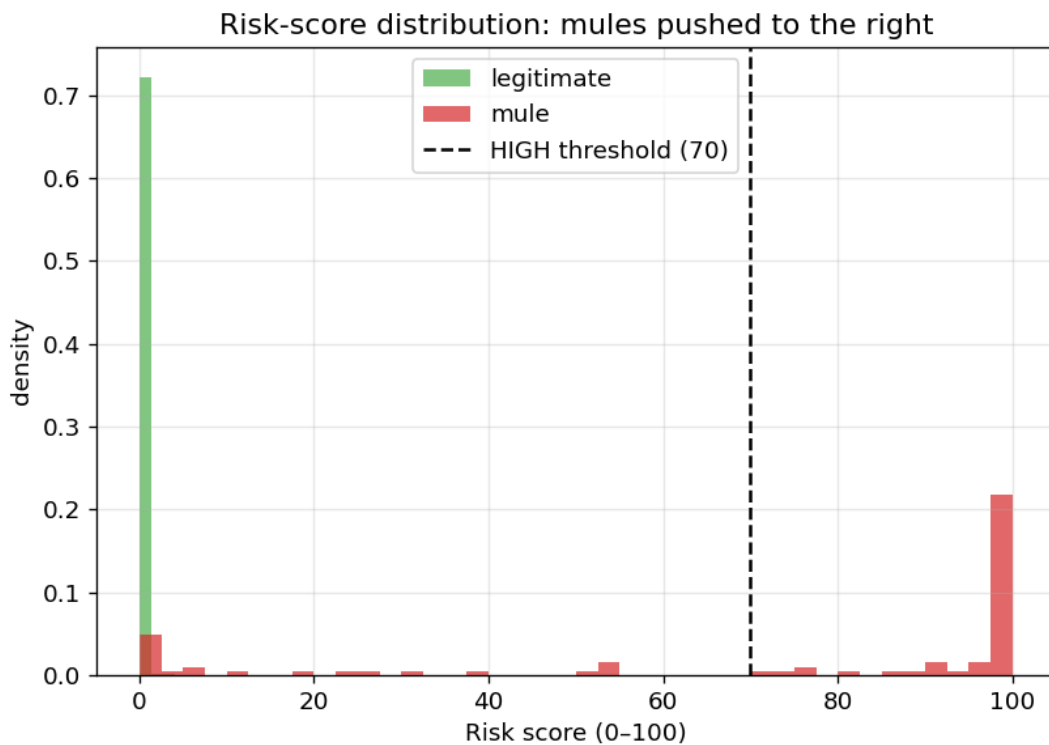


Fig 2. Risk-score distribution: legitimate accounts pile up near 0, mules at ~100 (with an honest hard tail).

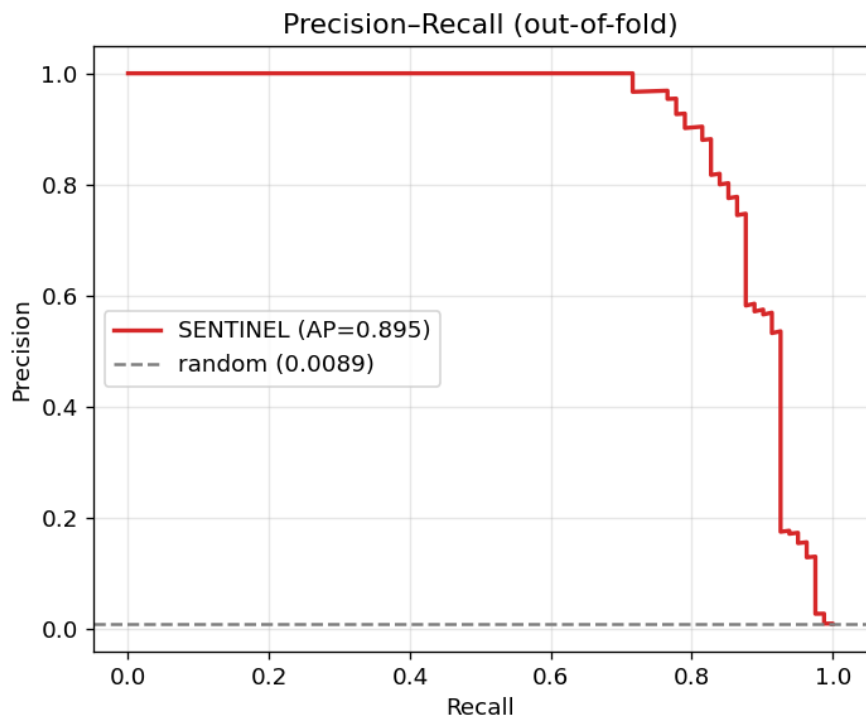


Fig 3. Precision-Recall (out-of-fold): perfect precision to ~70% recall, then the hard tail. AP=0.89.

5. Results (continued)

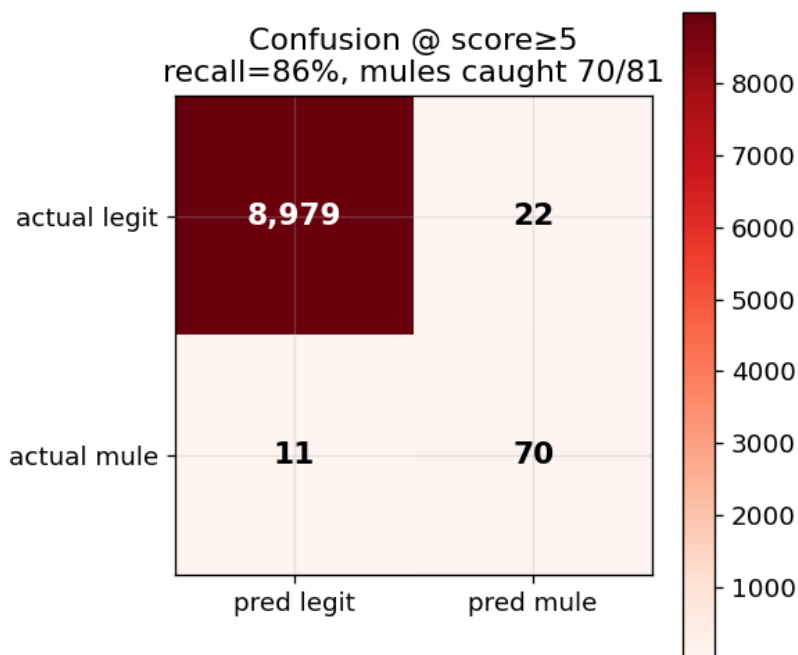


Fig 4. At a low operating threshold: 70/81 mules caught, only 22 false alarms across 9,001 legit accounts.

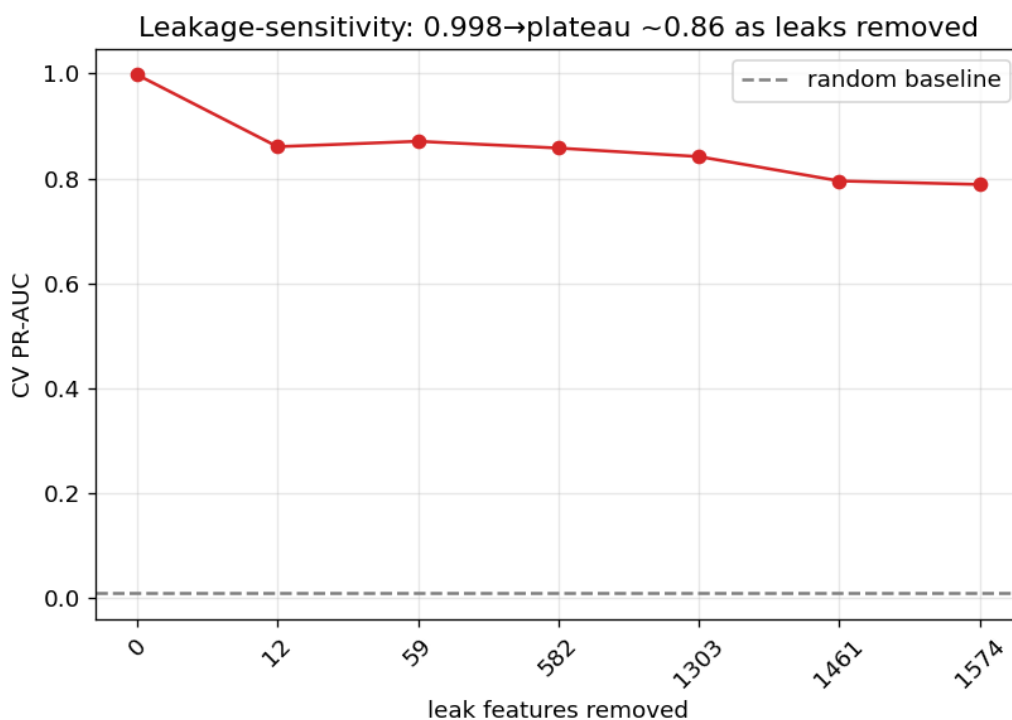


Fig 5. Leakage-sensitivity: score collapses 0.998 $\rightarrow$ ~ 0.86 as leaks are removed, then PLATEAUS - proving the remaining signal is genuine, not one hidden leak.

6. See It Work - Live System Output

These are REAL outputs from our running engine for mule account #9003 (not mockups) - the exact score, SHAP reasons, investigation report and API response an analyst/system receives.

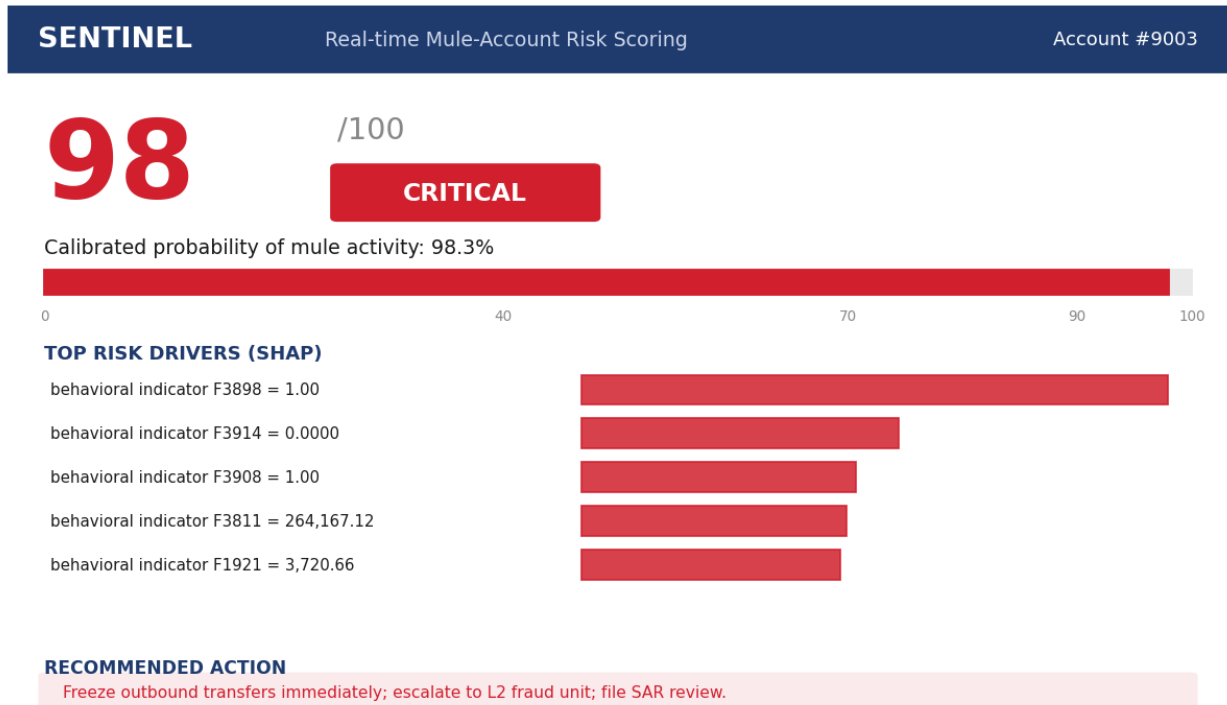


Fig 6. Real-time scoring card: a calibrated risk score, severity band, the SHAP drivers behind it, and a recommended action.

```
SENTINEL /score - real-time scoring API

$ curl -s -X POST localhost:8000/score \
  -H 'Content-Type: application/json' \
  -d '{"account_id": 9003, "features": {...} }'

{
  "account_id": 9003,
  "risk_score": 98,
  "band": "CRITICAL",
  "probability": 0.9831,
  "recommended action": "Freeze outbound transfers immediately; escalate to L2 fraud unit; file internal SAR review.",
  "top_drivers": [
    {
      "factor": "behavioral indicator F3898",
      "value": "1.00",
      "effect": "raises risk"
    },
    {
      "factor": "behavioral indicator F3914",
      "value": "0.0000",
      "effect": "raises risk"
    },
    {
      "factor": "behavioral indicator F3908",
      "value": "1.00",
      "effect": "raises risk"
    }
  ]
}
```

Fig 7. The live /score REST API response (FastAPI) - drop-in for any monitoring system.

6. See It Work (continued)

AUTO-GENERATED INVESTIGATION REPORT

INVESTIGATION REPORT – Account #9003

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Risk Score : 98/100 (CRITICAL)

Calibrated probability of mule activity : 98.3%

Recommended action : Freeze outbound transfers immediately; escalate to L2 fraud unit; file internal SAR review.

WHY THIS ACCOUNT WAS FLAGGED (top risk drivers):

1. behavioral indicator F3898 (anonymized): 1.00 – below the median; closer to the typical mule profile
2. behavioral indicator F3914 (anonymized): 0.0000 – in the bottom 5% of accounts; closer to the typical mule profile
3. behavioral indicator F3908 (anonymized): 1.00 – in the top 5% of accounts; closer to the typical mule profile
4. behavioral indicator F3811 (anonymized): 264,167.12 – below the median; closer to the typical mule profile
5. behavioral indicator F1921 (anonymized): 3,720.66 – below the median; closer to the typical mule profile

CONFIDENCE & CAVEATS:

- Score derived from 2233 populated features; 732 were blank for this account.
- Snapshot-based; not a substitute for transaction-level review.
- F3912 (post-hoc fraud flag) was excluded as leakage – this score reflects behavioral signal only.

Fig 8. The one-click, auto-generated investigation report - analyst-ready, with confidence caveats and the leakage note built in.

7. Business Impact & Intelligent Alerts

Rupee-cost decision engine (banks act on money, not PR-AUC)

Translating the threshold into money (assumptions, configurable: avg mule loss Rs 2,50,000; analyst review Rs 400; false-positive harm Rs 5,000): at a practical operating point we catch 85-89% of mules for a net ~Rs 1.7 crore saved per 9,082-account population. Net savings is flat across thresholds, so the operating point is analyst-capacity-bound - the bank chooses the recall/alert-volume trade-off it can staff.

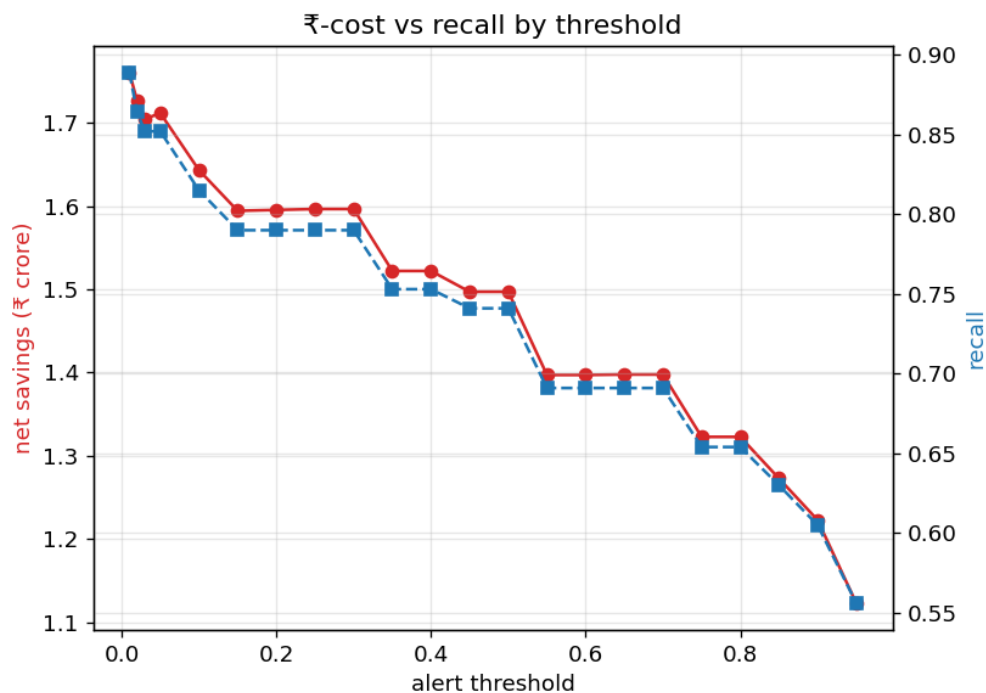


Fig 9. Net Rs-savings and recall vs alert threshold.

Intelligent, explainable alerts (PS2's 'intelligent alert generation')

Every alert carries: a 0-100 risk score + severity band (LOW/MEDIUM/HIGH/CRITICAL), the top SHAP-driven reasons in plain English, the account's deviation vs its peer cohort, a recommended action, and an auto-generated investigation report - so an analyst acts in minutes, not hours.

8. Feasibility, Prototype Roadmap & Honesty

Already working (reproducible)

- End-to-end pipeline (preprocess -> auditor -> model -> finalize -> insights -> scoring).
- Real-time FastAPI /score + /report service (~35 ms) and a Streamlit analyst dashboard.
- Deliverables: scored predictions for all 9,082 accounts + a ranked watchlist with reasons.
- Model card, smoke tests (5/5 passing), pinned requirements, and a self-contained Colab notebook.

Reproducible end-to-end - run it yourself. Repo: [GitHub link] | Live demo: [demo URL] | demo video: [link]

Phase-2 prototype plan (1 Jul - 17 Aug)

- GRAPH mule-RING detection (the standout upgrade): ingest shared device / IP / beneficiary links + circular fund-flow to catch networks, not just isolated accounts.
- Multi-feed fusion: live transaction streams + fraud / txn-monitoring alerts + govt cyber-fraud tickets + RBI/regulatory feeds, scored in real time to PREVENT CIRCULATION.
- Confirm with the bank which near-label features are pre- vs post-event (locks the exact number); plug in real cost figures for a per-channel cost-optimal threshold.
- Federated, privacy-preserving cross-bank intelligence; drift monitoring + scheduled leak re-audit.
- Hardening: streaming + batch scoring, alert-workflow / case-management integration, audit logging.

What we explicitly do NOT claim (maturity signal)

- Not a transaction stream (snapshot data) - we say so plainly.
- Anonymised features -> semantic labels in narratives are inferred, not bank-confirmed.
- 81 positives -> estimates carry real variance; we report a range and a confidence interval.
- Decision-support for analysts, not an autonomous freeze authority.

Bottom line: most teams will present a fake 1.000. We present a defensible 0.885, a 100%-precision watchlist, a rupee impact, and the leakage audit that proves we earned it.